Unit 2, Day 12

Doubling, Halving, and a Strategy Forum

Let's find out what happens to a product when we double or halve a factor.

Warm-up

Number Talk: Doubling and Halving

Find the value of each product. Be ready to share how you found it.

- 4×6

- 8×6

- 8×3

What happened to the product?

When one factor doubles, the product _____.

When one factor is cut in half, the product _____.

I know because _____.

Activity 1

Double One, Halve the Other

Solve each problem on your own. Then compare it to the problem before it. What changed? What stayed the same?

1. Find the value of each product.
   - 8×6
   - 4×12
   - 2×24
   - 1×48

2. Decide if each equation is true or false. Explain how you know.
   a. $4 \times 12 = 2 \times 24$

   b. $3 \times 14 = 6 \times 7$

   c. $5 \times 12 = 10 \times 24$

3. How could you change $5 \times 12 = 10 \times 24$ to make it true?

4. With your partner, say the big idea of this problem string in your own words. Write it here.

Activity 2

Maggie's Chickens: A Strategy Forum

1. Maggie's chickens live in a coop in her backyard. The coop is 12 feet by 13 feet. What is the area of the coop?

What is the problem asking us to do?	Reasonable estimate

Solve. Look for an efficient strategy, and show your thinking so someone else can follow it.

2. Find the value of 6×26.

What is the problem asking us to do?	Reasonable estimate

Solve. Then compare your answer to Problem 1. What do you notice?

3. After the forum: Which strategy was the most efficient for Problem 1? Finish the sentence.

I think _____ is the most efficient strategy, because . . .

_____

_____

_____

_____

Another way to say it:
The strategy with the fewest steps was _____. I know because _____.